PROJECT ARGUS

Military Value Proposition & Cost Analysis

Attritable HAPS Swarm for Hypersonic Cruise Missile Detection

1. The Detection Gap

The United States faces a critical and well-documented vulnerability in its missile defense architecture: the inability to reliably detect and track **hypersonic cruise missiles (HCMs)** during their terminal approach phase. Unlike ballistic missiles, which follow predictable trajectories detectable from space, HCMs fly at Mach 5+ at altitudes of 20-30 km — **below the optimal detection envelope of GEO satellites** like SBIRS, and **beyond the radar horizon of surface-based systems** like Aegis SPY-6 or THAAD AN/TPY-2.

This creates a **"detection gap"** — a band of altitude and range where current sensors cannot provide the persistent awareness needed for early warning. By the time a ship-based radar detects an inbound HCM at the horizon, reaction time may be as little as **15-30 seconds**. This is insufficient for any kinetic intercept response.

ARGUS fills this gap by operating **inside it** — persistent platforms at 60,000-70,000 ft (18-21 km), looking **upward** against the cold stratospheric background, where a Mach 5+ thermal signature is unmistakable.

2. Cost Comparison: Current vs. ARGUS

The cost differential between ARGUS and existing detection architectures is not incremental — it is **three to four orders of magnitude**.

System	Unit Cost	Coverage (km^2)	\$/km^2/day	Persistence
SBIRS GEO Satellite	$5.6 Billion	~hemisphere	$3,200*	24/7 (limited revisit)
HBTSS LEO Constellation	$14.4 Billion (28 sats)	~global	$1,800*	24/7 (constellation)
Aegis Destroyer (SPY-6)	$2.2 Billion	~160,000	$37,700	Patrol limited (fuel)
AN/TPY-2 THAAD Radar	$800 Million	~120,000	$18,300	Fixed/semi-mobile
E-7 Wedgetail AEW	$1.5 Billion	~300,000	$14,000	8-10 hr sorties
ARGUS (single unit)	**$4,800**	**~70,000**	**$0.19**	**Multi-day (solar)**
ARGUS (mil-hardened)	**~$50,000**	**~70,000**	**$1.96**	**Multi-day (solar)**

* Satellite \$/km^2/day estimated from program cost amortized over 15-year lifespan. ARGUS \$/km^2/day assumes 150 km detection radius (conservative SWIR + infrasound), unit replaced annually, 365-day operation.

For the cost of a single AN/TPY-2 THAAD radar ($800M), the United States could field 16,000 military-hardened ARGUS units — enough to blanket 1.12 billion km^2, or roughly three times the entire Pacific Ocean.

3. Theater Coverage: South China Sea Scenario

Scenario: Persistent hypersonic threat awareness over the entire South China Sea (3.5 million km^2).

Metric	Aegis Surface Group	ARGUS Constellation	Advantage
Units required	8 destroyers (4 on-station)	89 drones	Distributed
Capital cost	$17.6B	$14M	1218x cheaper
Annual operating	$400M/yr	~$5M/yr (replacements)	80x cheaper
Crew at risk	~2,640 sailors	0 (unmanned)	Zero risk
Single-point failure	One ship lost = 25% gap	One drone lost = 0.01% gap	Resilient
Detection mode	Active radar (detectable)	Passive (IR + acoustic)	Covert
Reaction time gain	Baseline (30s)	~5-10 min early warning	10-20x more time

Total ARGUS constellation cost for full SCS coverage: $14M vs. $17.6B for an equivalent Aegis surface group — a 1218x cost reduction with zero personnel at risk.

4. Adversary Cost Exchange Ratio

The attritable design philosophy creates an **asymmetric cost dilemma for adversaries**. To neutralize ARGUS coverage, an adversary must expend weapons against each drone individually. The cost exchange ratio is devastating:

Adversary Weapon	Weapon Cost	ARGUS Cost	Exchange Ratio	To Clear 100 Units
PL-15 AAM	~$2M	$50K	40:1 (our favor)	$200M vs $5M
HQ-9 SAM	~$6M	$50K	120:1 (our favor)	$600M vs $5M
S-400 / 40N6E	~$12M	$50K	240:1 (our favor)	$1.2B vs $5M
J-20 sortie (fuel+wear)	~$80K/hr	$50K	Pilot hours wasted	Logistics nightmare
Cyber/EW attack	Variable	$0 (mesh heals)	N/A	Swarm reconstitutes

Every ARGUS unit an adversary shoots down costs them 40-240x more than it costs the US to replace. This is the **Replicator thesis in action** — mass, low cost, and expendability as a strategic advantage.

5. Strategic Alignment: Why Now

DoD Replicator Initiative

Deputy Secretary of Defense Kathleen Hicks launched the Replicator initiative in August 2023 with the explicit goal of fielding "small, smart, cheap, and many" autonomous systems to counter mass. ARGUS is a textbook Replicator platform: **$4,800 per unit, fully autonomous, solar-sustained, and designed to be lost and replaced**.

JADC2 / ABMS Integration

ARGUS feeds directly into the Joint All-Domain Command and Control (JADC2) architecture. Each drone is a forward sensor node that generates track data and pushes it over LoRa mesh (swarm-local) and Iridium SBD (global uplink) to any JADC2-connected command center. The swarm acts as a **distributed sensor network**, not a single point of failure.

Passive Sensing = Covert Sensing

Unlike Aegis radar or AWACS, ARGUS emits no detectable radiation. The sensor suite is entirely passive — infrasound microphones listen for shockwaves, SWIR cameras detect thermal signatures. The platform itself is a small composite glider at 65,000 ft with no jet engine — effectively **invisible to adversary SIGINT and difficult to detect on radar**. It operates in contested airspace where active emitters would be targeted.

Persistent Forward Presence Without Risk

ARGUS provides what carrier strike groups and destroyers provide — forward presence and awareness — without putting 300+ sailors per ship at risk. In a conflict scenario, losing 50 ARGUS drones costs $2.5M and zero lives. Losing one destroyer costs $2.2B and potentially 330 lives.

6. Technical Novelty Assessment

Innovation	Prior Art	ARGUS Advancement
HAPS for missile defense	None — HAPS used for comms relay (Zephyr, PHASA-35, Loon)	First application of HAPS as a missile defense sensor layer
Passive infrasound at altitude	Ground-based CTBTO arrays (IMS network, fixed installations)	Airborne infrasound at 65 kft provides upward-looking geometry

Tiered sensor fusion cascade	Multi-sensor fusion exists (F-35 DAS, Aegis)	Power-gated cascade unique to solar-constrained platforms
Attritable HAPS swarm	Attritable UAS (XQ-58, CCA) but jet-powered, short endurance	Solar-sustained = days/weeks vs hours for jet attritable UAS
SWIR upward-looking from stratosphere	Satellite downward-looking IR (SBIRS/HBTSS)	Cold stratospheric background gives superior contrast ratio
Distributed acoustic TDOA across swarm	Single-platform acoustic (sonobuoys, ground arrays)	Multi-platform baseline of 10s of km enables precise geolocation

7. Platform Versatility

ARGUS is not a single-mission system. The same airframe, ground infrastructure, and manufacturing line support multiple mission sets with payload swaps:

Mission	Payload Config	Customer
Hypersonic missile detection	Infrasound + SWIR + EO (baseline)	MDA / INDOPACOM
Maritime ISR	EO/IR + AIS receiver	Navy / Coast Guard
Communications relay	SDR radio relay payload	Army / SOCOM
Persistent SIGINT	Wideband RF receiver	NSA / CYBERCOM
Atmospheric research	Radiosonde + lidar	NOAA / NASA
Border surveillance	EO/IR + radar	CBP / DHS
Disaster response	Comms relay + EO	FEMA / National Guard
Nuclear test monitoring	Infrasound + radionuclide	CTBTO / DTRA

8. Bottom Line

Metric	Value
Unit cost (COTS prototype)	$4,800
Unit cost (mil-hardened est.)	~$50,000
Coverage per unit	~70,000 km^2
Cost per defended km^2/day	$0.19 - $1.96
Endurance	Multi-day (solar-sustained)
Crew at risk	Zero
Cost to cover SCS (3.5M km^2)	**~$14.5M (vs $17.6B Aegis equivalent)**
Cost exchange ratio vs adversary	**40:1 to 240:1 in US favor**
Novel?	**Yes — first HAPS swarm for missile defense**

ARGUS is not an incremental improvement to missile defense — it is a paradigm shift. It replaces billion-dollar exquisite platforms with thousands of expendable sensors that cost less than a used car, operate for days without fuel, put zero personnel at risk, and create an asymmetric cost dilemma that no adversary can win.